

Reader's Guide



7月 19, 2026

Reader's Guide

How to Read *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields*

Purpose of This Guide

This monograph spans a wide range of mathematical disciplines, from variational calculus and differential geometry to topology, information theory, tensor networks, operator methods, and higher category theory. Consequently, readers from different backgrounds may find different reading paths more suitable to their interests and expertise.

The purpose of this guide is to suggest several recommended pathways through the text.

Path A — For Mathematicians

Readers whose primary interests are pure or applied mathematics may wish to emphasize the formal development of the theory.

Recommended reading order:

1. Mathematical Foundations
2. Variational Principles
3. Relational Field Equations
4. Symmetry and Conservation Laws
5. Differential Geometry
6. Gauge Theory
7. Topology
8. Information Geometry
9. Category Theory
10. Higher Structures
11. Mathematical Axioms
12. Appendices L–V

This path focuses on internal mathematical consistency.

Path B – For Physicists

Readers familiar with theoretical physics may prefer to compare SQFT with existing physical formalisms.

Suggested topics include

- Lagrangian mechanics,
- Hamiltonian formalism,
- gauge symmetry,
- renormalization,
- path integrals,
- effective field theory,
- tensor networks.

Throughout the text, it is important to remember that these concepts are employed as **mathematical analogies** rather than literal physical descriptions of social systems.

Path C – For Computer Scientists

Researchers interested in computation may concentrate on

- tensor representations,
- optimization,
- information theory,
- graph methods,
- machine learning,
- computational complexity,
- numerical simulation.

Many mathematical structures developed in SQFT naturally connect with modern computational methods.

Path D – For Researchers in Complex Systems

Readers working in network science, systems biology, economics, organizational science, or related disciplines may find the following progression useful:

1. Introduction
2. Relational Fields
3. Dynamical Systems

4. Information Geometry
5. Effective Theories
6. Tensor Networks
7. Applications discussed in later appendices

The emphasis should be placed on mathematical modeling rather than physical interpretation.

Path E – For Philosophers of Science

Readers interested in conceptual foundations may focus upon

- axiomatization,
- structural realism,
- mathematical modeling,
- scientific explanation,
- category theory,
- philosophy of mathematics.

Appendices O, P, V, and the Afterword provide a natural entry point for these topics.

Prerequisite Knowledge

The mathematical prerequisites vary according to the reader's objectives.

Essential Background

- Linear Algebra
- Multivariable Calculus
- Ordinary Differential Equations

Recommended Background

- Partial Differential Equations
- Differential Geometry
- Functional Analysis
- Probability Theory

Advanced Topics

Some later chapters assume familiarity with

- Lie Groups
- Fiber Bundles
- Algebraic Topology
- Category Theory

- Homological Algebra
- Operator Algebras

Readers encountering these subjects for the first time may consult standard references before attempting the corresponding chapters.

Reading Strategy

This book is **not** intended to be read as a conventional textbook in which every chapter depends strictly upon the previous one.

Instead,

the work is organized as a layered mathematical framework.

The recommended progression is

Foundations →Dynamics
 →Geometry
 →Information
 →Higher Mathematics
 → Applications and Extensions.

Foundations

→Dynamics→Geometry→Information→Higher Mathematics→Applications and Extensions.

Readers may enter the framework at different points depending on their interests.

Notational Conventions

Throughout the monograph,

- bold symbols denote vectors or structured mathematical objects where appropriate,
- uppercase calligraphic letters denote spaces or categories,
- lowercase Greek letters typically denote fields or parameters,
- uppercase Greek letters denote operators or global quantities,
- tensor indices follow standard Einstein summation conventions unless otherwise indicated.

Definitions, theorems, propositions, lemmas, remarks, and conjectures are numbered independently within each chapter.

A Final Recommendation

The framework presented in this volume is intentionally ambitious in scope.

Readers are encouraged to approach it in the same manner that one studies any broad mathematical theory:

- verify every proof independently;
- question every assumption;
- compare the framework with existing mathematical literature;
- distinguish carefully between established results and conjectures;
- and consider how the ideas may be refined, simplified, or generalized.

The value of a mathematical framework ultimately lies not in its breadth, but in the clarity and rigor with which its central ideas can be developed.



輸入留言

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Foundational Charter of the Social Quantum Field Theory Research Community

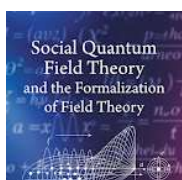
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The SQFT Charter Foundational Charter of the Social Quantum Field Theory Research Community Preamble The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry ...

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社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社 ...

量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響

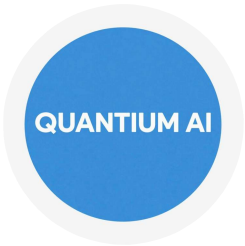
9月 27, 2025



量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響 摘要 在 2035年，全球局勢緊張之際，台灣的量子導航技術成為重塑國際權力格局的關鍵力量。本篇科幻論文式敘事，記錄了國家中山科學研 ...

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封存



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